Bootstrap Estimation of System Integrated Information

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Overview

Introduction to Consciousness

- Consciousness remains one of the most profound open questions in modern Science.

- "I think, therefore I am"
  - René Descartes (1641).

- How does the mind create conscious experience?

Figure: René Descartes

- Over 350 years later, we revisit consciousness from the perspective of a neuroscientist. What does consciousness physically arise from?

- **Integrated Information Theory (IIT)** Tononi (2004), is a leading theory that uses a mathematical framework to explain and measure consciousness (Φ).

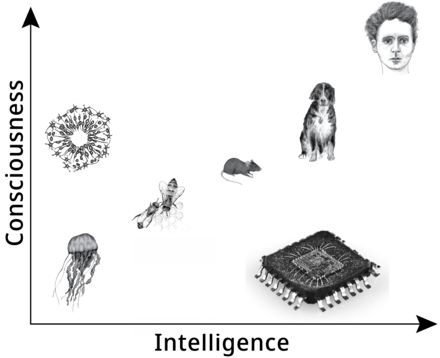


Figure: Findlay, Marshall, et, al., (2025)

- Under IIT, consciousness is attributed to the measure Φ: Integrated Information. This measure acts on a system and quantifies how its units acts as a unified whole, rather than independently of each other.

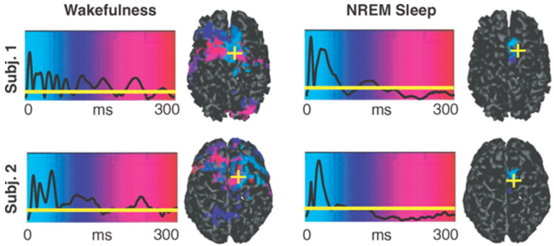


Figure: Massimini et al., (2005)

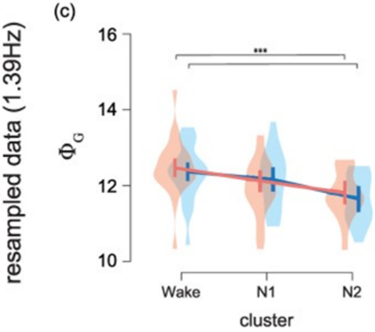


Figure: Onoda & Akama, (2024)

- We consider a system of interacting units in the form of a Markov chain.

- The system's current state, together with the node interactions specifies information about subsequent states.

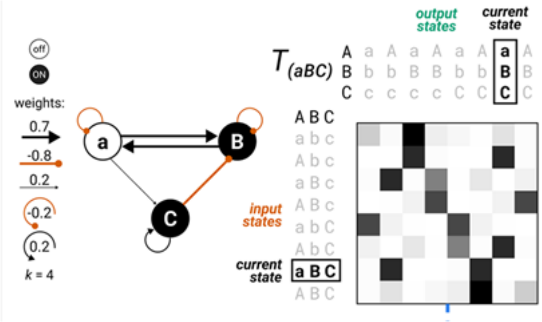
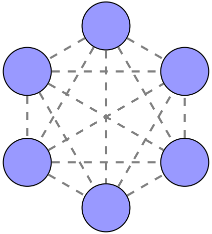
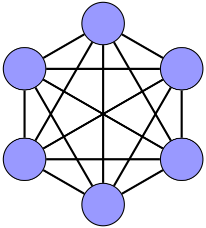


Figure: (Albantakis et al., 2023)

Integrated and Segregated Systems

- The information specified by the system about its subsequent states can be one two forms:



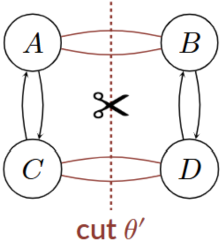


- Information specified by independent unit states.

- **Integrated Information** specified by the interactivity of units.

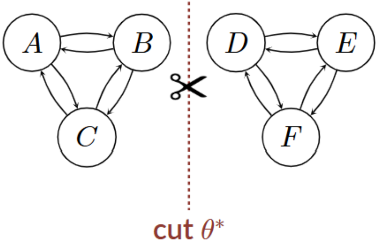
- To measure the amount of Integrated Information using Φ, we consider various cuts to the system and quantify the information lost.

Integrated ($\Phi > 0$)



Cut removes significant information

Segregated ($\Phi = 0$)

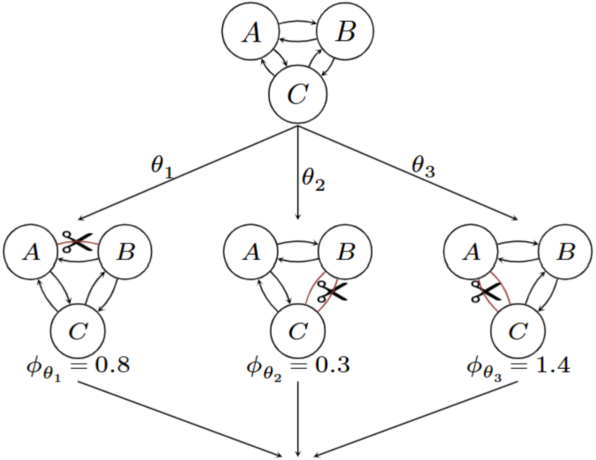


Cut removes no information

System $\mathcal{S}$

Cuts/Partitions θ_i

Information Loss $\Phi_{\theta i}$



$$\Phi(\mathcal{S}) = \min_i\{\phi_i\} = 0.3$$

The Computational Challenge

- To evaluate Φ we must locate the cut with minimal information loss. To perform this, we must exhaustively compute information loss for all possible cuts. For large-scale systems, this becomes computationally intractable.

$$M(5) = 1\ 061 \qquad M(7) = 61\ 888 \qquad M(9) = 4\ 419\ 572$$

Theorem (Number of Possible System Partitions)

For a fully connected system consisting of $\mathcal{N}$ units, the number of possible system cuts is

$$M(\mathcal{N}) \sim \mathcal{O}\left(\left[\frac{\mathcal{N}}{\log \mathcal{N}}\right]^{\mathcal{N}}\right)$$

Heuristic Measures of Integrated Information

- Heuristic measures have been proposed (Nilsen et, al., 2019) which consider a fixed subset of all possible cuts.
- Uncertainty quantification methods for these heuristic methods do not exist.
- Our work considered a randomized subset of cuts, where uncertainty quantification becomes possible.

Methodology Overview

Estimating Φ

Information loss values:

$$\Theta = \bigcup_{i=1...M} \{\phi_i\}$$

Random subset:

$$\mathcal{A} \subseteq \Theta, \quad |\mathcal{A}| = N$$

Φ Estimate:

$$\hat{\Phi} = \min_i \{\phi_i \in \mathcal{A}\}$$

$$\Longleftrightarrow$$

Estimating $X_{(1)}$

Information loss values:

$$\mathcal{X} = \{X_{(1)} \leq ... \leq X_{(M)}\}$$

Sample from $\mathcal{X}$

$$\mathcal{Y} = \{Y_{(1)} \leq ... \leq Y_{(N)}\}$$

Point Estimate:

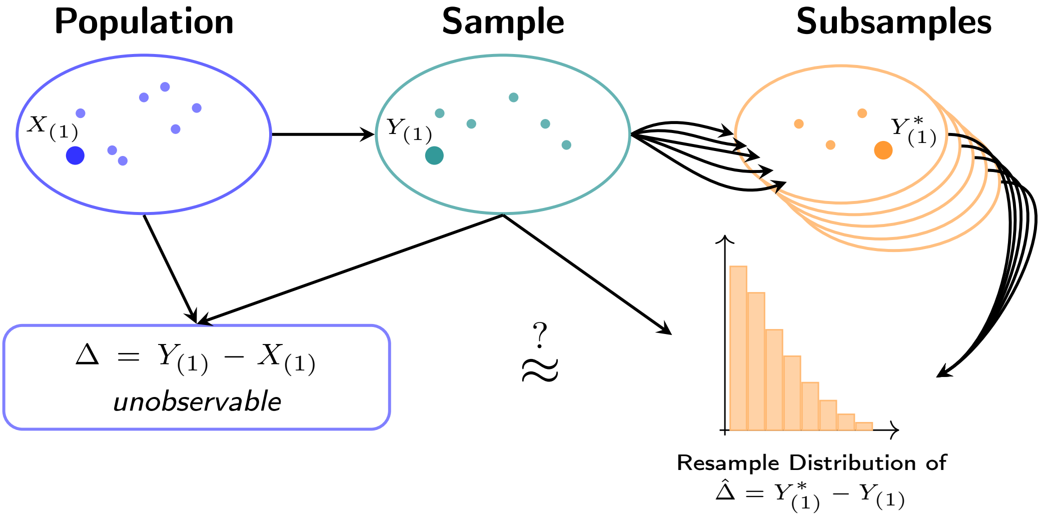
$$\widehat{\Phi} = \widehat{X_{(1)}} = Y_{(1)}$$

The Difficult Problem of Φ-Estimation

Estimating $\Phi = X_{(1)}$ is a uniquely difficult problem. Start by representing $Y_{(1)} \dots Y_{(N)}$ as a sample from a Discrete Uniform distribution over the set $X_{(1)} \dots X_{(M)}$.

1. Discrete EVT is formulated for distributions with integer-valued support.
2. Model fits using continuous EVT lead to degenerate distributions.
3. Bootstrap methods are typically inconsistent for endpoint estimation. (Bickel & Freedman, (1981).

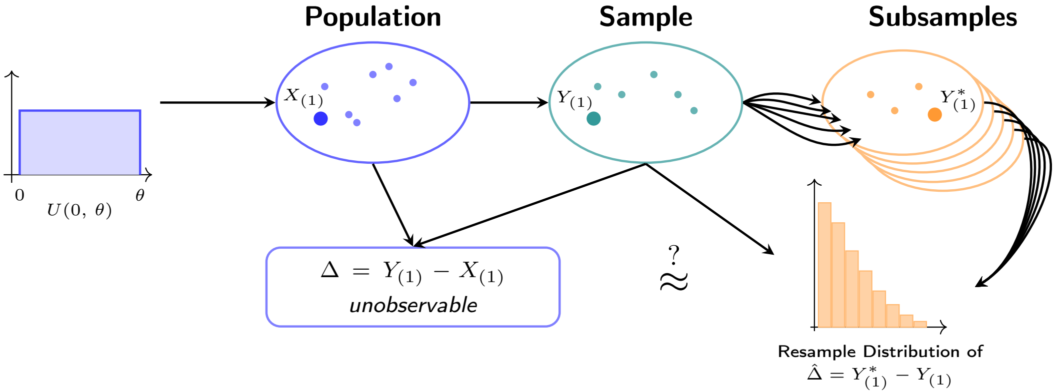
The Gap Bootstrap



The Gap Bootstrap

- In general, the method on the previous slide is inconsistent.
- Gap sizes between order statistics grow with smaller sample size. This effect is amplified in sparse regions such a distribution tails.
- To correct this, we will consider a contrived scenario in which the gap sizes are proportionate across sample sizes

The Gap Bootstrap Under Uniformity



Consistency of the Gap Bootstrap Under Population Uniformity

Theorem (Consistency of the Gap Bootstrap Under Population Uniformity)

Under population uniformity $X_1 \dots X_M \sim$ Uniform$(0, \theta)$, we draw subsamples of the same proportion as we take our sample (size N). Then as $M \to \infty$:

$$\hat{\Delta} = \frac{N+1}{M + \frac{N}{M}} \left(Y^*_{(1)} - Y_{(1)} \right) \xrightarrow{d} Y_{(1)} - X_{(1)}$$

- Thus if we can assume uniformity, the gap bootstrap is consistent.

The Bootstrap Stabilizing Transform: EVT Decomposition

- The assumption of uniformity in general does not hold. We instead assume the population $X_1 \dots X_N$ is generated from a distribution in the Weibull min-Domain of Attraction.

- Using this result, we can transform the lower tail of F_X to be locally Uniform

Definition (Weibull min-Domain of Attraction)

A distribution F is in the Weibull min-Domain of Attraction if there exists $\lambda > 0$ and $\alpha > 0$ such that

$$F(x) \sim \left(\frac{x-L}{\lambda}\right)^{\alpha} \quad \text{as } x \downarrow L.$$

where L is the lower support bound.

The Bootstrap Stabilizing Transform

- Under the previous assumptions, in the lower tail of the super-population cdf F_X, we have

$$F(x) \approx \left(\frac{x - L}{\lambda}\right)^{\alpha}$$

.

- Define the transformation $T : X \to W$ of the lower tail of the random variable X, defined by $W = (X - L)^{\alpha}$.

$$F_W(w) = \Pr(W \leq w) = F(L + w^{1/\alpha}) \approx \frac{w}{\lambda^{\alpha}} \iff W \sim \text{Uniform}(0, \lambda^{\alpha})$$

Main Result

Theorem (Consistency of the Gap Bootstrap Under Local Uniformity)

Let $\mathcal{X} = X_1 \dots X_M \sim$ Uniform$(0, \theta)$, locally in the lower tail and $\mathcal{Y} = Y_1 \dots Y_N$ be sampled without replacement from $\mathcal{X}$, with resamples $\mathcal{Y}^*_{(b)}$, $b = 1 \dots B$.

$$\widehat{Y_{(1)} - X_{(1)}} = \frac{N+1}{M + \frac{N}{M}}(Y^*_{(1)} - Y_{(1)})$$

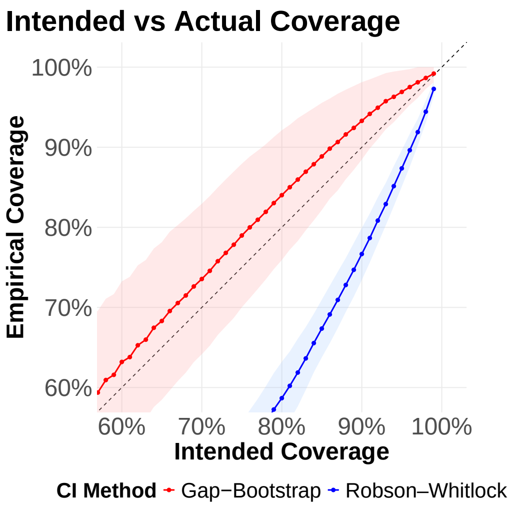
converges in distribution to the random variable $Y_{(1)} - X_{(1)}$ for large M.

- Therefore, by consistently estimating the transformation to uniformity, performing the Gap-Bootstrap under Uniformity, and transforming back to X, confidence intervals arise.

Simulation

- We generated $T = 50$ Simulated networks of 7 nodes ($M = 61888$) of type $\Phi > 0$, and fixed $p = N/M = 0.05$.
- The bootstrap stabilizing transform is estimated consistently using Hill's Estimator.
- For sake of benchmarking, we compare to the Robson $\&$ Whitlock's (1964) method for truncation-point confidence interval.

Simulation Results:



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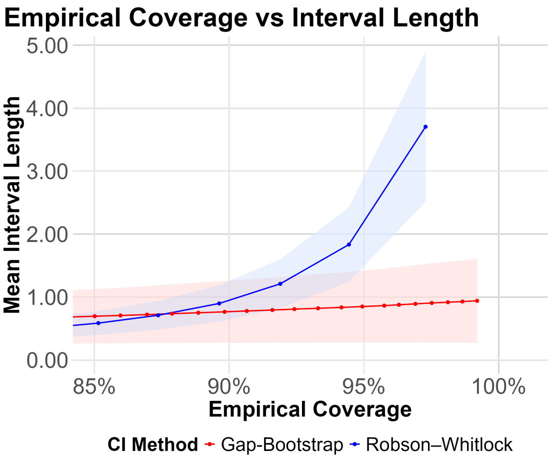

Discussion

- This work is the first method of interval estimation for Φ. This method is consistent under easily verifiable assumptions. The interval estimates attain their nominal coverage, and have shorter interval lengths than Robson & Whitlock's method.

- Future work includes the correction for the point mass found when $Y^*_{(1)} = Y_{(1)}$ with probability p. Also, proper analysis of the Hypothesis test $\Phi = 0$ defined from this interval.

- Applicability to other disciplines.

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